Clinical Practice Guideline

Management of Patients with Suspected or Actual Acute Spinal Injury

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to managing patients with suspected or actual acute spinal injury at Sir Charles Gairdner Hospital. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Note: Osborne Park Hospital (OPH) is not equipped to manage acute spinal injuries, please refer to Osborne Park Hospital Wide Guideline Suspected Spinal Injury – Investigation and Management

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Contents

1. General Instructions

1.1 Medical Requirements

If any clinical injury is detected or suspected, treatment plans including clear positioning descriptions must be documented in the inpatient notes.

All treatment plans must clearly indicate the name of the senior medical staff member with whom the registrar consulted, along with the Registrar's details.

After review and final report of the appropriate spinal imaging, by a consultant radiologist or senior radiology registrar, documentation of cervical spine clearance is required by **any one** of the following medical staff:

- Neurosurgical Consultant
- Intensive Care Consultant
- Emergency Department Consultant / delegated Registrar
- Orthopaedic Consultant / delegated Registrar
- Spinal Consultant (RPH)

1.2 Safety Information

The aim of care is to prevent further injury until injuries are confirmed or cleared.

Spinal clearance and / or confirmation of injury and management should occur as soon as possible, to reduce the incidence of complications associated with immobilisation.

If a cervical injury is suspected bolster and taping is required at a minimum until a definitive treatment plan is obtained from RPH Spinal Consultants or spinal clearance has been obtained and documented.

Full spinal precautions are required for all suspected or actual spinal injuries until otherwise documented.

If patient combative, collar not tolerated or unable to secure a proper fit, use of alternative supports (e.g. /bolsters, taping) in place of a cervical collar may need to be considered for patient safety and compliance.

Patients requiring full spinal precautions must be cared for on a Standard Hospital Foam mattress[1, 2]. The use of alternating air cell mattresses and low pressure/air loss mattresses is contraindicated.[1]

1.3 Nursing Management

- All nursing staff responsible for the care of patients requiring spinal precautions must have completed the Spinal Transfers for Nurses and HSA Employees course through NMAHS Education and Development Centre. Any staff member assuming responsibility for head holding during spinal repositioning **must** have completed this training.[2,3]
- Position restrictions for trauma patients with **suspected** spinal injury and awaiting clearance include:
  - Nurse in a supine or lateral position maintaining neutral spinal alignment as authorised.[4] The use of bolster / pillows for support is required to maintain an anatomical position if nursing patient in the lateral position.
  - Use of a cervical collar with suspected cervical injury[4]

- Head holding during re positioning / movement and if cervical collar is removed (e.g. for pressure care area)
- Log rolling is required until thoracolumbar [4] spinal clearance has occurred.
- Mobilisation in collar when thoraco-lumbar spinal clearance has occurred e.g. Semi-recumbent positioning[4]

1.4 Procedural Information

1. Transfer from ED to the ward will occur on the ED trolley with a nurse escort.
2. An ED HSA will accompany the patient and assist ward and ED nurse with trolley to bed transfer.
3. Slide board transfer is to be used to transfer the patient onto the bed.

2. Emergency Department Management (ED)

2.1 Medical Requirements

Directives for the management of such patients or clearance from spinal precautions can only be made by the patient's Consultant/ Registrar and must be documented in the patient integrated notes.[1]

2.2 Safety Information

All patients in ED are assessed utilising a prioritising sequence such as ABCDE; Airway with inline spinal immobilisation or precautions, breathing, circulation, disability exposure and environment.[5]

Patients with a suspicious mechanism of injury may have a pattern of symptoms indicative of suspected spinal injury.[1] Known trauma patients will have a trauma call activated and be assessed collaboratively with a Medical Officer (MO).

All patients with a suspected spinal fracture at any level must be assessed and treated as unstable until proven otherwise.[1]

Transferring from Ambulance stretcher to ED Trolley:

Full spinal precautions (FSP) are used when managing any suspected or confirmed unstable spinal injured patient ensuring[1]:

- Patient lying flat
- appropriate spinal immobilisation adjuncts in situ and patient educated not to move head, neck or back,
- Ensure the maintenance of spinal alignment, at all times.
- Ensure ED trolley is 'locked' to prevent accidental sitting up of patient with the trolley control.

For some patients' effective spinal immobilisation is vital to prevent cord damage yet excessive use of this practice can expose patients to the risk of additional complications such as iatrogenic pain, skin ulceration, aspiration and respiratory compromise.[6]

Refer to section 4 – Positioning Procedures and guidelines: Log Rolling.

In patients with a fixed flexion deformity e.g. ankylosing spondylitis (AS), spinal precautions need to be tailored to maintain the patient's pre-injury kyphotic curvature to prevent hyperextension.[7]

This can be achieved by the use of bolsters, pillows and foam/rolled blankets[1].
Scoop stretcher can be used to transfer patient across to ED trolley[1].

Use the log roll method to place slide board and slide sheet in situ, then transfer maintaining FSP[1].

2.3 Nursing Management

Note: *A minimum of 4 staff are required for this procedure (one must be experienced in cervical neck holds).[1,3]*

More staff may be required to manage lines, splints, slide board, etc.[1]

Document all transfers and methods used[1] in the Nursing Assessment Record.

Place lateral bolsters (e.g. firmly rolled and taped towels) either side of the head to prevent lateral movements and potential worsening of the spinal injury[1].

fit patient with a cervical collar (Philadelphia® Collar) as soon as possible, if appropriate[1].

Inform the patient/NOK about the importance of keeping the head and neck still.[1]

Assessment
On admission, patients with suspected spinal injury will be assessed by medical or nursing staff for the following[1, 8-9]**:**

Mechanism of injury/illness

- Level of spinal injury
- Neurological deficit
- Altered mental state
- Associated injuries e.g. limb, chest, arterial, abdominal or brain injuries
- Respiratory/cardiovascular function
- Co-morbidities e.g. diabetes, cardiac abnormalities, epilepsy etc.
- Level of intoxication as appropriate
- Pregnancy –refer to Obstetrics as soon as possible as minor alterations in maternal haemodynamic status may have profound ramifications for the foetus[1]

Management

- Clothing is removed[1]
- Excess linen is removed, and linen is smooth under the patient[1]
- No items (debris, personal effects) are under the patient[1]
- Document turns and pressure injury care[1] in the Nursing Assessment Record.
- Perform skin assessment[9]
- Ensure the patient is log rolled every 3 hours if clinically stable to reduce the risk of pressure injuries developing[9]

Note: *Log roll may be detrimental if haemodynamically unstable.[1]*

Transferring the patient from ED:
Nurse (+/- MO) experienced in spinal injury management to escort patient.[1]
Ensure the HSAs are aware the patient has a spinal injury.[1] ED HSA to escort patient, Extra care must be taken when pushing over uneven surfaces e.g. into lifts.[1]

3. General Nursing Management

3.1 Medical Requirements

Use of Graduated Compression Stocking (GCS) / Intermittent Pneumatic Compression (IPC) should be considered.[10] Low Molecular Weight Heparin (LMWH) may be used as thrombo-prophylaxis in the acute-care phase following Spinal Cord Injury (SCI) once there is no evidence or risk of active bleeding.[10] Refer to SCGH VTE Policy, refer to Clinical Practice Guideline– Venous Thromboembolism Prophylaxis.

3.2 Safety Information

Frequency of the following observations is determined by patients' clinical condition in liaison with Medical Team. Document and report any abnormal findings.

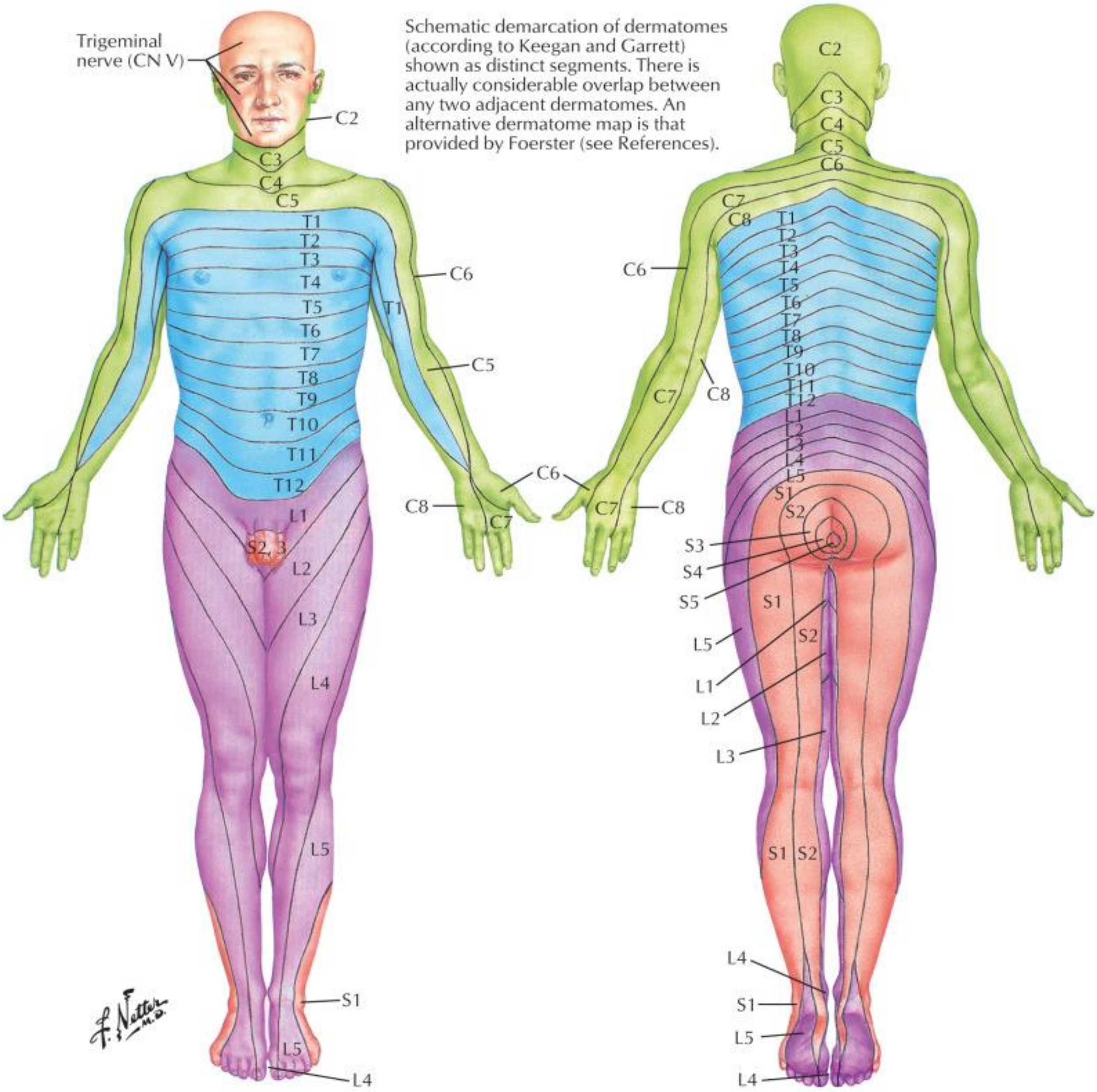
3.3 Nursing Management

- All patients with neurological deficit or complex fractures should have transfer to Royal Perth Hospital considered at the earliest safe opportunity.

Please refer to tables overleaf for managing:

- Suspected spinal injured patients with no neurological deficit until cleared.
- Actual Spinal injured patients with neurological dysfunction (first 48-72 hours).

Suspected Spinal Injured Patient with No Neurological Deficit Until Cleared	
Nursing Care	**Nursing Intervention**
Pain Control	Assess & document pain score[1,5,9] 4/24 and PRN. Regular analgesia prn.[1,5,9] Antispasmodics may be used if pain is related to spasms. Refer to Acute Pain Team.[1]
Observations	Core physiological observations hourly for the first 24hrs then 4/24 and prn as indicated by ORC Chart. Document respiratory rate, rhythm and effort in clinical notes.[1]
Neurological Assessment	Full neurological Observations are to be documented as baseline, and then performed as ordered, and PRN as indicated (for example, if the patient reports alteration in neurological function, or following procedures).[1] Dermatome assessment must be done as baseline and as required.[1]
Neurovascular Assessment	Monitor the neurovascular status of all four limbs as a baseline, and as ordered. If the patient has a complete SCI do not assess motor and sensory function frequently as this may cause patient distress.[1]
Accurate Fluid Balance & Bladder Care	Monitor bladder scans 4hourly – IMC if required as per Bladder Management Policy IV fluids should be considered if NBM. Indwelling Urinary Catheter (IDC) should be removed as soon as possible with intermittent catheterisation as indicated by bladder scanning to reduce risk of incidence of UTI.[1,11]
Oxygen Therapy	As per Oxygen Therapy CPG and Observation and Response Chart (ORC) chart. Encourage the patient with deep breathing and coughing exercises 1/24.[1]
Bowel Management	Monitor bowel sounds 4/24[1] for first 48hrs. Report abdominal pain and distention to treating team. Patient at risk of Paralytic Ileus.[12] In the absence of Ileus, commence a bowel regime to avoid constipation as per treating medical team.[1,12]
Oral Intake	Assess bowel sounds 4/24.[1] Patient NIL BY MOUTH until medical review after bowel sounds have returned.[1,12] Treatment with a PPI should be considered[1] as patient at high risk of Gastric Ulcer.[9]
Pressure Area Care	Complete Waterlow risk assessment on admission. Nurse patient on a firm mattress in the supine neutral position to maintain Spinal Alignment; until otherwise documented by medical staff. Pressure area care to be provided 2/24,[1,4,9,13] with collar care 4/24 until mobile.[1,4] Consider use of prophylactic pressure 'off-loading' dressings i.e. Mepilex™ sacrum/ heel
VTE Prophylaxis	Intermittent Pneumatic Compression / Graduated Compression Stockings should be considered, [10] medical assessments as per SCGOPHCG VTE Prophylaxis CPG.
Provide care as per table *Suspected Spinal Injured Patient with No Neurological Deficit Until Cleared* (**above) with the addition of:**	
Oxygen Therapy	Forced vital capacity readings to be measured as a baseline and as requested by treating team.[1]
Education	Patient and family education and support should be commenced immediately. Communication, Psychological support, family involvement and education needs should be considered.[5] Allied health staff should be involved in the patient's care.



Levels of principal dermatomes

C5	Level of clavicles	**T10**	Level of umbilicus
C5, 6	Lateral sides of upper limbs	**L1**	Inguinal region
C8, T1	Medial sides of upper limbs	**L1, 2, 3, 4**	Anterior and medial surfaces of lower limbs
C6	Digit I (thumb)	**L4, 5, S1**	Foot
C6, 7, 8	Hand	**L4**	Medial side digit I (great toe)
C8	Digits IV and V (ring and little fingers)	**L5, S1, 2**	Lateral and posterior surfaces of lower limbs
T4	Level of nipples	**S1**	Lateral margin of foot and digit V (little toe)
		S2, 3, 4	Perineum

Figure 1: Dermatome map[14]

4. Positioning Guidelines and Procedures

4.1 Medical Requirements

Authorisation and documentation from any one of the following medical staff is required prior to any manoeuvre or positioning other than supine:

- Neurosurgical Consultant
- Intensive Care Consultant
- Emergency Department Consultant / delegated Registrar
- Orthopaedic Consultant / delegated Registrar
- Spinal Consultant RPH.

4.2 Safety Information

The main aims of care for trauma patients with suspected spinal injuries are:

- All patients should be nursed in a neutral position and maintaining spinal alignment unless the Orthopaedic Registrar or Spinal Consultant has requested and documented the patient's cervical spine to rest in hyper flexion, flexion, neutral, extension or hyperextension. This must be clearly documented.
- Prevention of possible further spinal injury,
- Application and correct fitting of cervical collar,
- Instigation of spinal position restrictions,
- Prevention of complications of immobilisation,
- Collar care,
- Pressure area care,
- Timely completion and reporting of radiological procedures.

Patients with unstable injuries must never be placed on airflow or dynamic alternating mattresses unless documented by the consultant. HoverMatt® and HoverJacks® are **NOT** to be used for a spinal injured patient.[1]

4.3 Nursing Management

Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

The person maintaining head hold is responsible for:

- Checking documentation to ensure planned move is approved.
- Performing the cervical head hold and communicating with the patient.
- Delegating roles and informing all team members of the exact procedure.
- Ensuring all equipment required is immediately available, bed brakes are on and safe handling principles are maintained by all team members.
- Directs all steps of the procedure and counts out all steps to ensure smooth moves.
- Ensures that full neutral spinal alignment is maintained throughout the move and once complete.

Please refer to Section 5. Patient Handling Options for information regarding patient equipment options.

<u>Spinal alignment:</u>

Supine:

Centre of forehead, chin, xiphoid sternum and pubic bone in line centrally. The shoulders and pelvic crests should not be rotated. Refer to Figure 2.

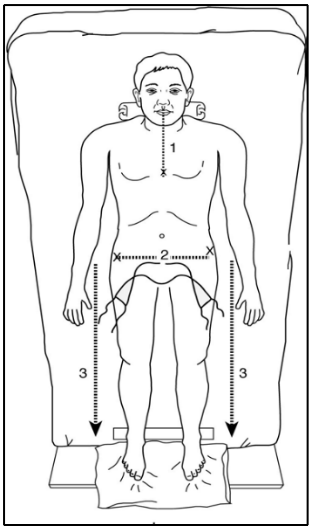


Figure 2: Supine position with no pillow under head (Image source: unknown)

Lateral:
Patient must be at 90 degrees to the bed, and the ear, iliac crest and external malleoli are in alignment.[1] Patient should be in neutral spinal alignment at all times.

- The WHOLE BED should be routinely tilted head up 30 degrees for all patients at risk of aspiration (e.g. receiving enteral feeds, intubated patients, patients with compromised airway). This may be contraindicated in patients with tetraplegia as diaphragmatic pressure may be required to support spontaneous breathing. Consult with Orthopaedic team regarding risk versus benefit of this positioning.
- Patients with neurological deficit should have pillows placed at the end of the bed to ensure both feet are flexed at 90-degree angles.[1] Refer to Figure 3 below.

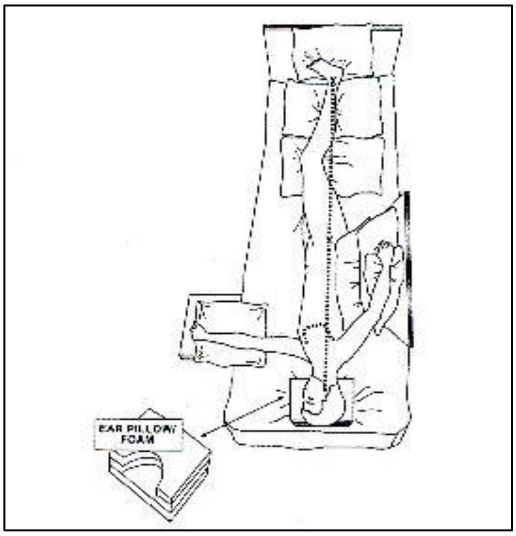


Figure 3: Lateral position (Image source: unknown)

Cervical Head Hold:
Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

Prior to spinal clearance, the patient's head and neck must be supported during position changes, including during any procedure for which the collar is removed, to ensure neutral spinal alignment is maintained at all times.

Once the cervical spine is documented as cleared, head holding is no longer required[4] but may be necessary to assist with comfort for the patient. However, until the patient's thoraco-lumbar spine is cleared, care must be taken to ensure that the patient's head remains in neutral spinal alignment on turning and lateral positioning.

1. Explain the procedure to the patient regardless of conscious state and ask the patient to lie still and to refrain from assisting.[4]
2. Ensure that the collar is correctly sized and fitted prior to commencement.[4]
3. If applicable, ensure that devices such as indwelling catheters, intercostal catheters, ventilator tubing etc. are repositioned to prevent over-extension and possible dislodgement during repositioning.[4]
4. Appropriate Personal Protective Equipment (PPE) should be worn.

There are two recommended methods of head holding[4]:

1. Head holding from the top of the bed is the recommended method in ED[4] and ward environments.

2. Head holding from the side may be useful in Intensive Care where equipment restricts access to the top of the bed and airway management is paramount.
3. The designated head holder stands at the head or side of the bed with the bed at a comfortable height i.e. above waist height.[4]
4. The head hold must be in place prior to directing other team members to commence their role and is not removed until the procedure is otherwise complete, patient alignment checked, and all supporting padding is in place.

Method 1: For Head hold from the top of the bed:

1. Support the side of the patients' head with forearms. Remove any bolsters.
2. Place thumbs over the clavicles to gain stability.
3. Cross fingertips at lower cervical area. See Figure 4 below.

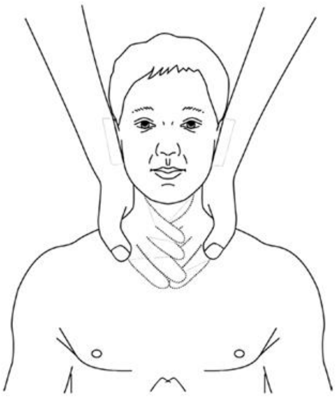


Figure 4: Head hold from top of bed[15]

Method 2: For Head hold from the side of the bed[4] (appropriate for log rolls): (For intubated patients only)

1. Head holder stands to the side of the bed towards which the patient will be rolled.
2. Place one hand under the patient's neck with fingers spread.
3. Place other hand over the patient's jaw (for a ventilated patient, the ETT may be supported with the thumb and index finger).
4. Apply firm pressure to restrict flexion, extension and tilting. See Figure 5 below

Figure 5: Head hold from side of bed[4]

Log Roll

Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

A log rolling procedure is required prior to thoraco-lumbar spinal clearance[4]. If the cervical spine is not cleared, but the Thoraco-lumbar spine is cleared, the patient must be rolled with a cervical head hold in place.

Minimum staff required for Log roll[3]:

- Cervical +/- Thoraco-lumbar injury = minimum of <u>4 staff</u>
- Thoraco-lumbar injury only = minimum of <u>3 staff</u>

<u>Note:</u> The log roll can be utilised to facilitate examination of the patient's back, cervical collar care, pressure care, and to position slide sheets/ spinal boards.

Ensure to[4]:

1. Explain the procedure to the patient regardless of conscious state and ask the patient to lie still and to refrain from assisting.
2. Ensure the collar is correctly fitted, prior to commencement of procedure.
3. Devices such as indwelling catheters, intercostal catheters, ventilator tubing etc. are repositioned to prevent overextension and possible dislodgement during repositioning.
4. If the patient is intubated or has a tracheostomy tube, airway suctioning prior to log rolling is suggested, to prevent coughing which may cause possible anatomical misalignment during the log rolling procedure.
5. Position the bed at a suitable height for the head holder and assistants.
6. Prior to commencement of the log rolling procedure, the patient must be supine and anatomically aligned.
7. The patient's proximal arm must be adducted slightly to avoid rolling onto monitoring devices e.g. arterial or peripheral intravenous lines. The patient's distal arm should be extended in alignment with the thorax and abdomen or bent over the patient's chest if appropriate. (Refer to Figure 3)
8. A pillow should be placed between the patient's legs.
9. The assistant supporting the patient's upper body places one hand over the patient's shoulder to support the posterior chest area, and the other hand around the patient's hips.
10. The assistant supporting the patient's abdomen and lower limbs overlaps one hand around the patients' hips above the other assistant, and the other hand over the patient's thighs.
11. On direction from the head holder the patient is turned in neutral spinal alignment in one smooth action.

On completion of the planned activity, the head holder will direct the assistants to either return the patient to the supine position or to support the patient in a lateral position with wedge/spinal pillows. The patient must be left in neutral spinal alignment, unless positioning in flexion/ extension prescribed and documented.

Lateral Positioning

Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

The patient may be positioned laterally prior to spinal clearance, to assist with chest physiotherapy and reduction of collar-related occipital pressure.[4]

Exceptions may include unstable thoracic, lumbar or pelvic fractures. In this case, clarification of position restrictions needs to be obtained and documented from the treating unit/team.[4]

The patient must be well supported in the lateral position using wedges/spinal pillows. The patient's head and body must be kept in neutral spinal alignment at all times.[4]

Padding may be required between the cervical collar and the bed to prevent lateral tilting of the patient's head.[4]

Supine to Lateral Positioning

The person closest to the patient's head is always responsible for the count and coordinates the move.

1. Explain the procedure to the patient.
2. Remove (as necessary) the arm boards and foot roll/heal elevators and ensure any lines will not be compromised by moving the patient
3. Hold the patient by placing hands
   a. 1st - person on the shoulder and hip
   b. 2nd - person on the hip and thigh
4. On the count of three roll the patient on to their side and place the slide sheet under the patient.
5. On the count of three to roll the patient back onto their back and straighten the slide sheet.
6. Carefully remove the neck roll.
7. On the count of three to move the patient across the bed toward the team.
8. Place the axilla pillow insitu for the patient to roll onto.
9. Place the ear pillow on the bed as close to the patient's head as possible with cut out facing the top of the bed.
10. On the count of three to use the slide sheet to position the patient on their side.
11. Remove the slide sheet carefully.
12. Adjust the ear pillow so is fits snugly against the patient shoulder and so the ear fits within the cavity. Ensure the ear is not crushed.
13. Position back support cushions and pillows.

Ensure:

- No two skin surfaces come into contact by placing pillows between the patient's legs:
  - The uppermost leg is flexed at the hip and knee to prevent the patient rolling back onto their back
  - The lower leg is positioned straight, and a foot roll is placed under the lateral malleolus.
  - Pillows are placed at the end of the bed so lowest foot is flexed at 90 degrees
- The arm closest to the bed is well supported by placing an arm board and pillows in position (refer to Figure 3)
- If the shoulder is too far under the patient's body, ensure an assistant supports the scapula area to gently pull the patients shoulder through
- Ensure the patient has the call bell and other necessary items before leaving.

Lateral to Supine Positioning

<u>The person responsible for the head hold is always responsible for the count and coordinates the move.[3]</u>

1. Explain the procedure to the patient.
2. Remove (as necessary) the arm board, back support and pillows, foot roll and ensure any lines will not be compromised by moving the patient.
3. Hold the patient in position by placing hands:
   a. 1st person on the shoulder and hip
   b. 2nd person on the hip and thigh
4. Place the slide sheet in position

5. On the count of three roll the patient on to their back
6. Roll the patient to the opposite side as above to ensure slide sheet is in place
7. On the count of three to move the patient to the centre of the bed
8. Direct a team member to carefully remove the ear pillow
9. Straighten the slide sheet
10. Place the head pillow on the bed as close to its intended position as possible
11. Remove the slide sheet
12. Carefully place neck roll in position if required.
13. Position arm boards and pillows
14. Place heel elevators in position
15. Place pillows at the end of the bed to ensure both feet are flexed at 90 degrees
16. Place bolsters on either side of the head to deter lateral rotation
17. Ensure the patient has the call bell and other necessary items before leaving
18. A touch bell may be necessary if patient has any sensory or movement deficit

Cervical Injuries Equipment

- Cervical collar
- Bolsters
- Appropriately sized posture/support pillows as required, which may include Neck roll
  - Head pillow
  - Ear pillow
  - Arm boards
  - Heel elevators
  - Pillows to support limb (approximately 7)

Document the size and position of the pillow/s required to achieve the desired position in the inpatient notes.

4.5 Procedural Information

Information integrated in to nursing management as above.

5. Patient Handling Options

5.1 Safety Information

- Patients' neutral spinal alignment must be maintained when using the Jordan / Scoop frame, slide sheets, and slide board/spinal board.
- Care must be taken during application of the Jordan Frame or Scoop to avoid causing skin tears.
- HoverMatt® and HoverJacks® are NOT to be used for a spinal injured patient.[1]

5.2 Nursing Management

A Jordan frame is used for changing linen, transferring patients from the bed to a trolley, bariatric patients, patients with multiple fractures and patients with pelvic fractures.

5.3 Procedural Information

The following equipment is available to assist in the positioning of patients. Refer to Section 4 for Positioning Guidelines and Procedures.

Jordan Frame / Scoop Frame

A Scoop or Jordan Frame along with a hoist may be utilised as an alternative to a slide board to manoeuvre patients from one bed to another.

Neutral spinal alignment and a spinal head hold must be maintained during application and removal of the frame.

A minimum of 4 people is required[1] (1 head holder, 2 to place frame / slats and 1 to stabilise chest/operate hoist/provide nursing care).

Jordan Frames are available in ICU / G53 / Hospital Equipment Store
Scoop Frames are available in ED
Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

Information integrated in to nursing management as above.
1. Head Holder commences Cervical Spine hold and directs removal of bolster.
2. The head holder then directs positioning of frame, and slats if Jordan Frame in use. Slats are placed with the broadest slat at the patients' head. 1 person should provide stability to the patient's chest/hips during slat positioning to avoid rotation.
3. The patient is then repositioned, and all nursing care required is provided with the cervical hold remaining in place.
4. The head holder then directs removal of the slats (again ensuring chest/hips are stabilised) and frame.
5. The head holder ensures correct neutral spinal alignment and directs any adjustments necessary.
6. Bolsters are replaced and cervical hold removed.

Slide sheets
Slide sheets can be used to manoeuvre supine patients either up / down, or left / right in a bed, and from one bed / trolley to another. The patient must be log rolled to place the slide sheet.

A spinal head hold and neutral body alignment must be maintained during all movement, unless positioning in flexion/ extension prescribed and documented. A <u>minimum of 4 people</u> are required.

Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]
1. The head holder must be experienced in the use of slide sheets and direct the team throughout the procedure.
2. Slide sheets must be removed from under patient once procedure is complete and can be done without rolling the patient, however counter traction to prevent patient movement may be required.

Slide board / Spinal board
Full length rigid slide boards may be placed under patients to facilitate sliding the patient from one bed/trolley to another.

The patient must be maintained in neutral spinal alignment, unless positioning in flexion/ extension prescribed and documented.

The head holder must maintain a spinal head hold for the duration of the procedure. A <u>minimum of 4 people</u> are required (1 /head holder, 2-3 to roll the pt, 1 to position board and assist with transfer).

Slide boards are available in all clinical areas.

Prior to repositioning ensure an experienced, appropriately trained person assumes head hold position to coordinate procedure.[2,3]

<u>The person holding the patient's head is always responsible for the cervical spine hold and coordinates the removal of bolsters.</u>

1. The head holder directs the team to log roll the patient (as per the log roll technique) using the sheet.
2. Position the slide board under the patient and sheet.
3. Direct the team to roll the patient back into the supine position and on to the slide board.
4. For the transfer, 2 -3 team members stand on the side that the patient will be transferred to with the second bed/trolley between them and the patient. Ensure all brakes are on.
5. Move patient in a 2-stage manoeuvre as instructed by the head holder. When instructed by the head holder team members grip the sheet close to the patient and pull the sheet so that the patient slides across the bed/trolley and onto the second bed/trolley. The team is then directed to perform a log roll to remove the board.
6. Once the patient is back in supine position, the head holder directs staff to remove their holds, checks spinal alignment. Bolsters are replaced and then release the cervical hold.

6. Cervical Collar Management

6.1 Medical Requirements

Documentation of cervical spine precautions or requirement for rigid collar in inpatient notes

6.2 Safety Information

Measuring for collars must only be completed by nursing, physio or medical staff experienced in the procedure.

- Correct size
- Correctly fitted

The Philadelphia® collar has been shown to exert a pressure, above capillary closing pressure, on the occiput, mandible and chin.[16]

Use of the collar for periods longer than 48-72 hours has been associated with increased risk of pressure injury.[4,17??]

A Miami J® collar should be considered for unconscious patients, if the patient has a confirmed cervical injury, a serious head injury, or is requiring inotropic therapy.

A Miami JTO® Thoracic extension should be considered for cervical and high thoracic injury.

Patients required to wear a collar for > 48 hours should be considered for a Miami J® collar.[18]

Philadelphia and Miami J Select collars are kept in G53, G52 & G66.

Miami JTO Thoracic extensions are kept in G53, G52, and G66Philadelphia collars are also in ED.

6.3 Nursing Management

- Collars and extensions must be checked a minimum of each shift to ensure that correct fit is maintained, especially in the setting of neck and facial swelling. Pressure points should be checked to minimise pressure injury.[19]
- Pillow should not be used under the patient's head unless documented by Medical Order.[19]

- Patients must be nursed on a standard hospital foam mattress.[1,2] The use of alternating air cell mattresses and low pressure/air loss mattresses is contraindicated.[1]
- Patients should be issued with two complete collar Philadelphia collars to allow for cleaning and drying in the first 48-hour period if not changed to Miami J collar.[19]

6.4 Procedural Information

1. The collar should be removed every 4 hours to inspect the neck for signs of pressure.[1,4]
2. The patient's head and neck must be held in neutral spinal alignment by another staff member until the cervical collar is replaced. If the patient is mobile once per shift is adequate.[1]
3. The Philadelphia® collar should be washed in warm, soapy water and dried with a towel each day.[4]
4. The Miami J® and JTO thoracic extension comes with a second lining. The lining should be washed in warm soapy water each day.[19]
5. Heavy soiling which is unable to be removed may necessitate a replacement collar.[1,4]
6. The patient's skin must be washed and dried thoroughly and inspected for signs of pressure.[4]
7. The collar must be replaced prior to log rolling for removal of the posterior section and inspection of the occiput.[4]
8. The patient's hair should be washed with the head held in neutral spinal alignment as a minimum weekly.[1] Matted or knotted hair or road grime beneath the collar may cause increased skin pressure, therefore hair must be combed or trimmed to prevent this occurrence. Hair may need to be clipped beneath the collar around wounds and to view the occiput.[1,4]

When a Miami JTO thoracic extension is in use, ensure:

- Skin hygiene is performed around the brace.
- The collar and the extension should be left insitu to shower or wash
- Patient and family education on the fitting and care of the brace is very important prior to the patient's discharge from hospital.

Choice of Collar

1. Philadelphia® (Adjustable) Tracheotomy Cervical Collar

The Philadelphia® Cervical Collar restricts cervical spine flexion, extension and rotation. Philadelphia® collars are Latex free, water resistant and x-ray, CT and MRI lucent.[20]

Collar Fitting Procedure

This is only to be completed by nursing, physio or medical staff experienced and skilled in the procedure. Position the patient supine, with the nurse, physio or medical staff providing cervical head hold throughout the procedure.

Always maintain the patient's head in neutral spinal alignment.

A second nurse, physio or medical staff member **MUST** maintain a **cervical head hold throughout** the process of fitting and applying a Philadelphia® collar.

To do this you will need two staff members:

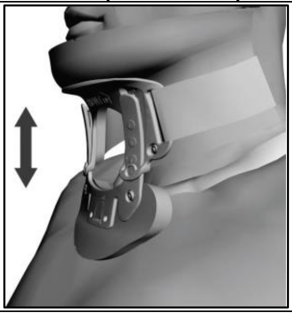
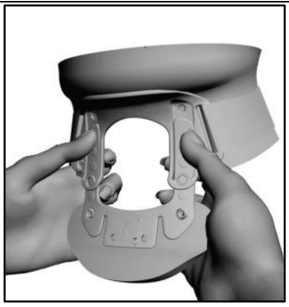
1. Maintain the cervical hold.
2. Fit and apply the collar.

Discard the separator between the FRONT and Sternum collar pieces.[20]

Philadelphia® (adjustable) collar sizing[20]

<u>Neck Height:</u> Place the chin-cup of the front to the patients chin and slide the lower front down to the patient´s sternum (Picture 1). Lift the front away from the patient and push the two tabs to lock the size (Picture 2).

<u>Circumference:</u> Identify your patient´s profile type according to the chart in (Picture 4). Adjust the circumference by tearing away sections of the back-piece foam to change to a smaller circumference (Picture 3).

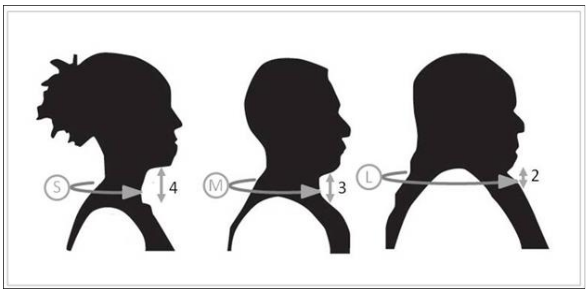
		
Picture 1	*Picture 2*	*Picture 3*

Collar Sizing Guide (Neck height and circumferential adjustment) [20]:

Table 1 below represents common sizing for the illustrated patient types (Picture 4).

Patient Type	Neck Height	Circumference	Circumference Adjustment
Stout	2	Large	No tear off
Average male	3	Medium	Tear off 1 section on each side
Thin female	4	Small	Tear off 2 sections on each side

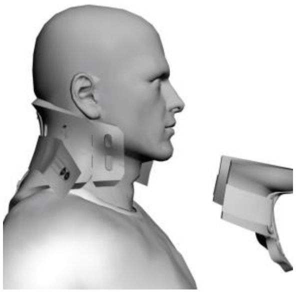
Table 1: Philadelphia® collar common sizing[20]



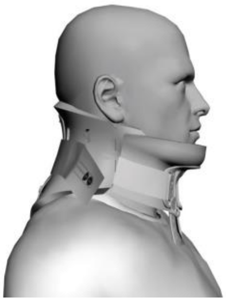
Picture 4: Philadelphia® collar: Illustration of patient types[20]

Philadelphia® (adjustable) Collar Application[20]

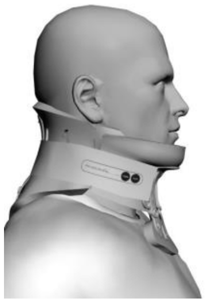
1. The two-piece design consists of a FRONT piece and a BACK piece which are packaged as a set. After adjusting the size of the collar, apply the back piece of the collar to the back of the patient’s neck. Centre the collar (Picture 5). The arrow on the back should point upward.
2. Apply the FRONT piece of the collar with the chin secured in the recess.
3. Centre the collar to secure neutral spinal alignment. The front piece OVERLAPS the back piece to ensure effective immobilization and comfort (Picture 6).
4. The arrow on the front should point upward. With hook and loop fasteners, tighten the collar with a bilateral adjustment
5. (Picture 7). This will secure the patient’s cervical region in neutral spinal alignment.
6. Ensure the collar is comfortable for the patient.
7. The nurse providing cervical head hold should check for neutral spinal alignment.



Picture 5



Picture 6



Picture 7

Additional information is available from: www.ossur.com.au

2. Miami J Select® Cervical collar

The Miami J Select® collars provide secure rigid support.

Aspen™ / Miami J Select® collars are indicated for high-risk groups of unconscious trauma patients who are not expected to be cleared of spinal injury within 48 hours of admission and patients with confirmed fractures requiring immobilisation for the management of their fracture.

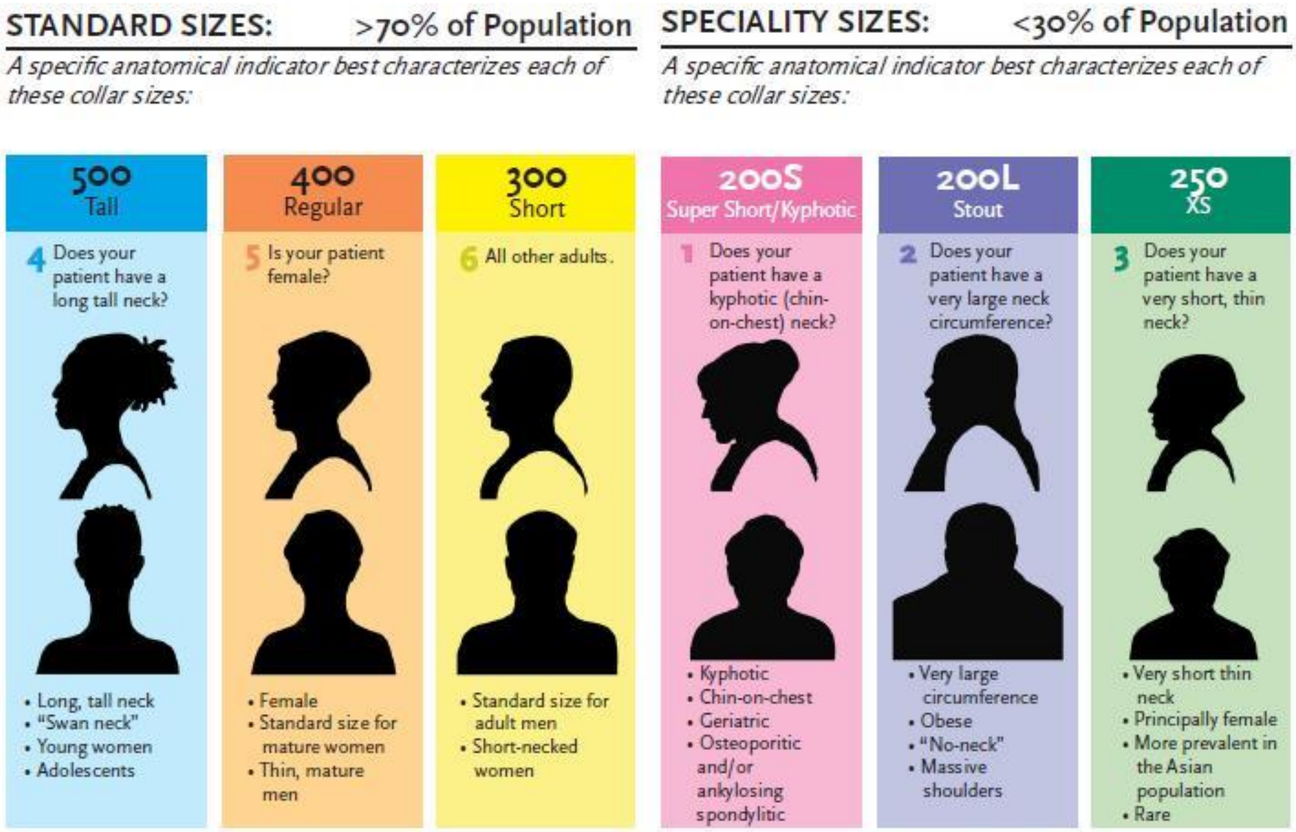
The spinal consultant should be consulted regarding choice of collar to ensure correct patient management.

Miami J Select® Collars are Latex free and MRI compatible.[21]

Measuring the patient

IMPORTANT: this is only to be completed by nursing, physio or medical staff experienced and skilled in the procedure.

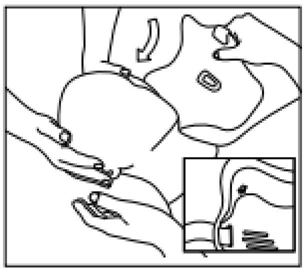
Miami J Select is an adjustable collar suitable for most adult and adolescent patients; some patient phenotypes and anatomical variations may not be accommodated. The Miami J Select is suitable for Stout, Tall, Regular, and Short patient phenotypes (picture 8). The Miami J Select has not been tested in Super Short and XSmall patient phenotypes and should therefore not be used on these patients.



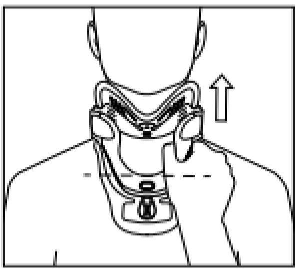
Picture 8 Miami J® collar sizing: Phenotypes[22]

Applying the Miami J Select® collar[21]

1. Cervical head hold is required throughout the procedure
2. Remove any pillows from behind the patient's head. Position patient with arms to the side, shoulders down and head centrally aligned.
3. Ensure the padding extends beyond the edge of the plastic and the Sternum Relief Knob is oriented vertically as shown in (picture 11).
4. Slide the Back Panel behind the patient's neck and centre it, noting arrow on panel pointing up (picture10). Note: Long hair should be placed outside of the plastic.
5. Slide the Front of collar up the chest wall until the bottom of the tracheal opening is at the level of sternal notch. Chin Support may not contact chin at this point (picture 11). Sides of the collar Front should be oriented up, off the trapezius, and toward the ears.
6. Holding the collar against the chest with the bottom of the tracheal opening at the level of the sternal notch, depress the Height Adjustment Button and manually adjust the Chin Support to achieve the desired height (picture 5).
7. When desired height is attained, release Height Adjustment Button to lock in place.
8. Verify that desired height is achieved on both sides of the device. Depress Height Adjustment Button and make micro-adjustments as necessary.
9. While holding the Front securely, place the sides of the collar Front within the sides of the Back Panel. Apply the Straps of the back panel to the Hook Landing Areas on the front. Tighten Straps alternately to an equal length on both sides. (picture 13)
10. If further height adjustment is required, depress Height Adjustment Button, and manually move to position. Use the Height Indicator Marks to record patient's collar height.
11. When desired position is attained, engage the Patient Compliance Lockout located behind the Height Adjustment Button by moving the lever to the left. (picture 14)
12. Straps must be aligned with the Hook Landing Areas. When patient is properly fit, there should be equal amounts of excess straps overhanging the Hook Landing Area. These may be trimmed (picture 15)

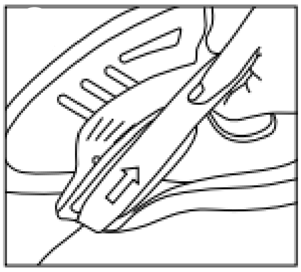


Picture 10

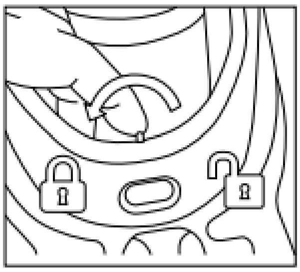


Picture 11

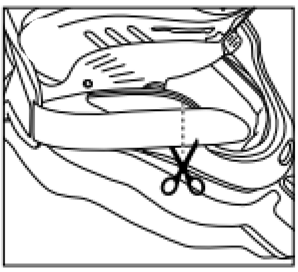
Picture 12



Picture 13



Picture 14



Picture 15

3. Miami JTO® Thoracic Extension

The Miami JTO® Thoracic Extension provides more control than a collar alone. This extension is designed to treat cervical and high thoracic injuries. The spinal consultant should be consulted regarding choice of collar and extension to ensure correct patient management.

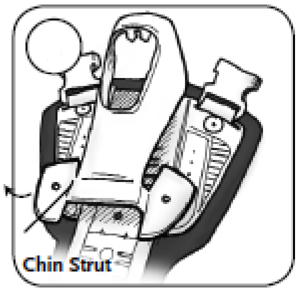
Extension fitting procedure
IMPORTANT: this is only to be completed by nursing, physio or medical staff experienced and skilled in the procedure.

Patients under C- Spine precautions require supine application of the Miami JTO®.[23]

Pre application of Miami JTO[23]

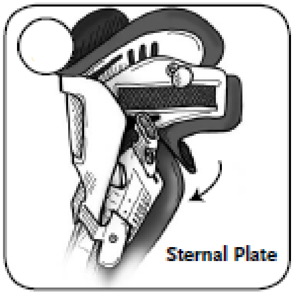
1. Remove blue and white sternal pad from collar (Picture 16.)
2. Remove chin strut from Miami JTO front piece (Picture 17.)
3. Squeeze collar to expose chin rivet. Attach the chin strut. The top opening of the chin strut will slide around the centre rivet on the collar and snap into place (Picture 18.)
4. Attach the chin strut feet to the chest plate in the lowest possible position (height may be adjusted later). The sternal plate on the collar attaches to Velcro at the top of the Miami JTO chest plate. The rivet holes in the chin strut can be aligned with corresponding holes on the chest plate to insure symmetrical positioning (Picture 19 & 20).
5. If not already attached, attach Miami JTO front to chest plate, using one of the two bottom openings. The Miami JTO front should be set at the highest possible position and should be above the costal margin (Picture 21.)

Picture 16

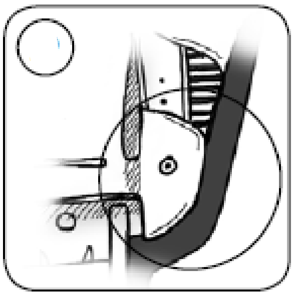


Picture 17

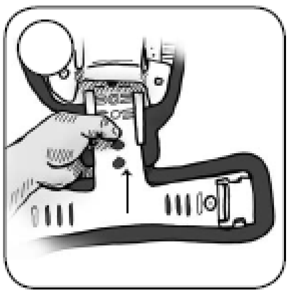
Picture 18



Picture 19



Picture 20

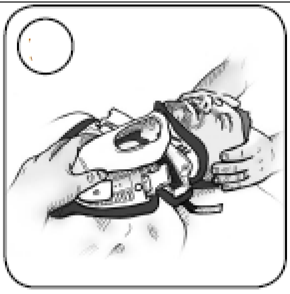


Picture 21

Miami JTO® application[23]
Patients under C-Spine precautions require supine application of the Miami JTO®.[23] Cervical head hold is required throughout the procedure

6. Ensure patients head is in position and stabilised using a head hold (Picture 22.). Centre the collar back piece on patients' neck and attach Velcro straps to front of collar. Tighten straps one at a time to an equal length on both sides. Velcro straps should be aligned "blue on blue".

7. Log roll patient onto side. Place the Miami JTO back piece on the bed under the patients' back, with shoulder and waist straps spread out (Picture 23.)
8. Return the patient using a log roll and head hold to the supine position (Picture 24.)
9. Attach the Miami JTO® front to the back using the shoulder buckles. Adjust the shoulder straps for a snug fit. Both shoulder straps should be of equal length and front and back pieces should be centred on the patient (Picture 25.)
10. Loosely buckle waist belts and pull even and tight. (Picture 26.)
11. Once patient is log rolled into upright position adjust height of the chin strut to fit comfortably under patient's chin (Picture 27.)
12. Adjust all the straps as necessary to properly fit. Once the Miami JTO® is in place you should be able to freely insert a finger behind the top of the chest piece. (Picture 28.)

		
Picture 22	Picture 23	Picture 24
		
Picture 25	Picture 26	Picture 27
		
Picture 28		

7. Jewett Brace Management

7.1 Medical Requirements

Documentation that a Jewett Brace is recommended must be clearly documented in inpatient notes, by medical staff.

7.2 Safety Information

An Orthotist will measure and fit the Jewett brace with the assistance of a physiotherapist.[24]

The Jewett Brace is designed to support the thoracic and lumbar spine, whilst maintaining alignment and preventing any twisting and bending forwards.[24]

A Jewett brace should always be fitted whilst lying down flat in bed.[24]

It is highly recommended that patients wear a light cotton singlet or t-shirt under the brace to avoid skin irritation.[24]

It is important to use the **log roll technique** when removing and reapplying the brace, and also whilst getting in and out of bed.[24]

7.3 Nursing Management

The Jewett brace must be checked a minimum of each shift to ensure that correct fit is maintained. Pressure points should be checked to minimise pressure injury.

The Jewett brace must be worn when showering, sitting, walking and travelling in a car.[24]

The brace may be removed when the patient is lying down flat and at night when the patient is sleeping (unless specified otherwise by the medical team).[24]

7.4 Procedural Information

Applying the Jewett Brace[24]

Picture 29

1. Position the patient on their back and place the Jewett Brace across their chest and abdomen (Picture 29.)
2. The spine support pad will be on the patients left side.
3. The bottom of the brace should sit just above the pelvic bones. The lock on the left side should stay unlocked.

Picture 30

4. Log roll the patient onto their right side and ensure the spine support pad is placed centrally over their back (Picture 30.)
5. Place your left hand behind their back and use it to push the strap under your right side, so that the strap is moved to the front.

Picture 31

6. Log roll back the patient on to their back.

7. Place the keyhole clip over the metal screw on the right side and lock it into place (Picture 31.)

Picture 32

8. Then, close the lock on the left side of the brace; this will tighten the brace (Picture 32.)

9. Listen for it to click down and be careful not to pinch your skin when closing it.
   The brace should fit firmly on their body.

Removing the Jewett Brace[24]

Picture 33

1. Lie the patient down on the bed on their back.

2. Unlock the Jewett Brace on the left side; this will loosen the brace (Picture 33)

3. Pull the keyhole clip up over the metal screw on the right side.

Picture 34

4. Log roll to the right to remove the spine support pad (Picture 34).

Picture 35

5. Log roll patient back onto their back.

6. Remove the brace from their chest and abdomen (Picture 35).

8. Transport to Another Institution

8.1 Medical Requirements

The name of the spinal consultant accepting care of the patient at the destination hospital must be clearly documented in inpatient notes.

Specific spinal management required for transfer must be clearly documented in the inpatient notes.
Adequate analgesia and antiemetic must be prescribed for transport.

8.2 Safety Information

Patients with acute spinal cord injury or unstable spinal fractures must be transported by paramedic ambulance with a nurse escort/s. The nurse escort must be experienced in caring for spinal patients and in cervical head hold.
Maintain spinal precautions / stabilisation throughout transport.

8.3 Nursing Management

- A nurse escort is always required for unstable fractures and patients with spinal cord injury.
- Assessment by ward / area CNS or After-hours CNS is required in determining the need for, and number of nurse escorts required. Two nurses (or a medical officer and a nurse) will be required when cervical precautions are in place.
- Ensure patient has been given antiemetic to reduce the incidence of vomiting during transport.
- Ensure adequate analgesia given prior to transport.
- Reassure and explain procedure to patient and significant others.
- Confirm availability of bed at destination prior to leaving the ward.
- Transport can be booked via St John's Ambulance. A taxi voucher should be provided for escorting staff returning to the hospital.
- Police escort is not required for transfer of patients with actual or suspected spinal injury.
- All unstable spinal fractures required a VAC Mat for transport.[5,25]

1. Procedural Information

1. Explain procedure to patient.
2. Transfer patient from bed to ambulance stretcher maintaining spinal precautions. Patient record or copies of relevant inpatient notes to be provided (check with ward / department clerk).
3. Provide clinical handover to receiving nursing staff and complete nursing transfer summary
4. If ambulance not returning to SCGH, escorting staff are to return by taxi.

Related Documents

Legislation:

- Work Health and Safety Act 2020
- Medicines and Poisons Act 2014 and the Medicines and Poisons Regulations 2016

Policy:

- SCGOPHCG Health Consent to Treatment Policy
- SCGOPHCG Infection Prevention and Control Manual – Hand Hygiene
- SCGOPHCG Clinical Handover - Communicating for Safety

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Research Reviewer:	A Folkard A/ CNC (Orthopaedics), K Guthrie (CNS Emergency Department) E Kenny (Acting Coordinator of Nursing Surgical Services)				
Applicable Standards	NSQHS Standards Std 1: Clinical Governance | Std 2: Partnering with Consumers | Std 3: Preventing and Controlling Healthcare Associated Infection | Std 5: Comprehensive Care | Std 6: Communicating for Safety | Std 8: Recognising and Responding to Acute Deterioration				
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